



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE

GRADE 12

CIVIL TECHNOLOGY: WOODWORKING

NOVEMBER 2024

MARKING GUIDELINES

MARKS: 200

These marking guidelines consist of 21 pages.

INSTRUCTIONS FOR MARKERS

1. Markers should:

- Familiarise themselves with the question and answer before evaluating the responses of candidates.
- Always interpret the responses of the candidates within the context of the question.
- Consider any relevant and acceptable answer during pre-marking but should strictly adhere to the answers after finalisation of the marking guideline.
- There are TWO approaches to answering questions; these are (1) to describe and (2) to explain.
 1. If a candidate is required to explain e.g., a process in 4 steps, only the first 4 responses should be considered.
 2. However, if for example candidate is required to explain or describe a process, we need to consider that that candidates may write a long description, not necessarily well organised. In this case the marker needs to evaluate the complete statement to judge if the candidate explained the required outcome satisfactorily and allocate marks on merit.
- Mark what the candidate wrote and do not interpret or predict responses.
- Indicate the tick or cross right at the position where the mark needs to be awarded or where the candidate made the error.
- Accept the letter corresponding with the correct answer as well as the answer written in full in multiple-choice questions or similar questions.
- Accept incorrect spelling in answers unless the spelling changes the meaning of the answer.
- If a learner writes two or more answers separated by a slash (/) mark only the first response, unless the additional answer/s are different names for the same item e.g., Yale lock/Night latch. In this case, the answer for the response should be awarded and the slash (/) should NOT be considered as an additional answer.

2. For calculations:

- A mark is only awarded if the correct unit is written next to the answer. If the question states that the answer must be in a specific unit, a mark will ONLY be awarded if the answer has the correct unit as indicated in the question.
- Marks will only be allocated for the correct values if the candidates add instead of multiply. NO marks will be awarded for the calculations and the answer.
- Where an incorrect answer is correctly carried over, the marker must recalculate the values, using the incorrect answer from the first calculation. If correctly used, the candidate should receive the full marks for subsequent calculations.
- Alternative methods of calculations must be considered, provided that the correct answer is obtained.
- For the calculation of quantities marks will be awarded for the correct use of the dimension paper.

3. When marking drawings:

- The member for which the mark should be awarded should be drawn correctly in the correct position to receive a mark.
- A member incorrectly drawn but wrongfully repeated in another position will be awarded the mark for the repeated incorrect member provided that the marking guideline provide for TWO or more marks for that member (positive marking).
- Marks can only be awarded for a label if the label is correctly indicating the correct member. Do not consider labels for members of which the labels were provided on the answer sheet.
- Scale drawings should always be marked using an appropriate mask.
- If the incorrect/wrong drawing was drawn, the candidate can be awarded for only what was provided for on the marking guideline.
- If a line diagram or an orthographic view instead of a pictorial drawing (isometric/oblique/perspective) is drawn, the first assessment criteria for each member will be marked wrong, but marks will be awarded for the subsequent members if TWO or more marks are awarded for the same member.
- If candidates draw/give more information than what is required, mark strictly according to the assessment criteria.

4. Incorrect numbering of questions:

- If a candidate numbered an incorrectly, but the answer is in the correct position according to the sequence of the questions in the question paper, circle then the incorrect numbering and mark the response.
- If questions were answered randomly not following the same sequence as in the question paper and the learner numbered incorrectly, the response should NOT be marked.

5. Duplication of responses and questions answered in the correct place:

- If a question is answered twice, mark the first response.
- If a question should be answered on an answer sheet and the candidate answered it on both the answer sheet and in the answer book, mark the response on the answer sheet and cancel the response in the answer book.
- If the question was answered in the answer book instead of on the answer sheet, mark the response in the answer book according to the assessment criteria on the marking guideline.

QUESTION 1: OHSA, MATERIALS, TOOLS, EQUIPMENT AND JOINING (GENERIC)

- 1.1 1.1.1 3 m ✓ (1)
- 1.1.2 3 m ✓ (1)
- 1.1.3 Fire-fighting measures ✓ (1)
- 1.1.4 ensure that the ladder does not protrude excessively ✓ (1)
- 1.1.5 Powder coating ✓ (1)
- 1.1.6 the thickness of under size parts can be increased ✓ (1)
- 1.1.7 head and pin ✓ (1)
- 1.1.8 preventing backing off ✓ (1)
- 1.1.9 wood preservatives ✓ (1)
- 1.1.10 spiral ✓ (1)
- 1.2 B ✓ (1)
- 1.3 Diameter of thread/Size of Rawl bolt ✓ (1)
- 1.4 Properties and advantages of cured concrete:
- 1.4.1 • Improves the protection of the steel reinforcement ✓
 • Improves the protection against rust
 ANY ONE OF THE ABOVE (1)
- 1.4.2 Cured concrete can carry more weight. ✓ (1)
- 1.5 A multi-detector can be used:
- to detect materials in or behind a wall/carpentry/plumbing/construction surface. ✓
 - to determine distances to and from an object.
- ANY ONE OF THE ABOVE**
- A laser level can be used:
- to indicate horizontal and/or vertical levels/heights/slopes. ✓ (2)

1.6 Uses of dumpy level:

- Determine differences between levels and vertical heights ✓
- Determine levels ✓
- Determine slopes
- Determine horizontal angles/angles
- Setting out of buildings
- Transferring levels and heights
- Determine the distance from the dumpy level to an object/telescopic staff

ANY TWO OF THE ABOVE

(2)

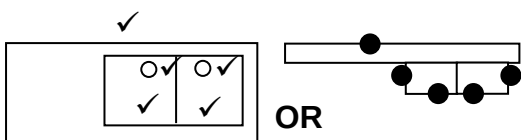
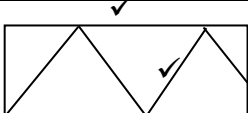
1.7 Caring for the telescopic staff:

- The staff must always be folded when it is transported ✓
- Keep it in a bag to prevent damage/scratching of the graduation during transportation ✓
- Ensure that the plastic or metal clips are in working order to keep the sections in position
- Clean after use
- Store in a safe dry place

ANY TWO OF THE ABOVE(2)
[20]

QUESTION 2: GRAPHICS AS MEANS OF COMMUNICATION (GENERIC)

NO.	QUESTIONS	ANSWERS	MARKS
1.	What is the measurement of the dwelling facing Hibiscus Street?	12 000 mm/12 m ✓	1
2.	Identify number 1.	Manhole/MH ✓	1
3.	Identify the number indicating the building line.	4 ✓	1
4.	Identify number 2.	Boundary line ✓	1
5.	Identify number 3.	Driveway/Entrance/Exit ✓	1
6.	Write down the abbreviation for number 5.	RE ✓	1
7.	Explain the purpose of number 6.	Garage door: To enter/close/secure the garage ✓	1
8.	How many fluorescent tubes are used in number 7?	1 x 40W /1/One ✓	1
9.	How many one-way switch single-pole switches are in the dwelling?	4/Four ✓	1
10.	How many built-in cupboards are in the dwelling?	2/Two ✓	1
11.	What is omitted in terms of electrical installation in the bedrooms?	Switch socket outlets/Socket outlets/Wall plugs ✓	1
12.	Name TWO access points into the dwelling.	Access points: • Sliding door/Lounge door/Door at Number 8/Number 8 ✓ • Garage doors/Rollup door/Door at Number 6/Number 6 ✓	2
13.	How do you know in which direction the door at number 8 will open?	By the direction of an arrow ✓	1
14.	Name ONE other fixture that can be installed in the bathroom indicated by number 9.	Bath/Bidet/Urinal/Vanity/Bathroom cabinet/Wash hand basin/Basin/Extractor fan/Heated towel rail ✓ (Abbreviations is NOT accepted)	1
15.	Identify number 10.	Hot-water cylinder ✓	1

16.	Which plot is on the eastern side of plot number 23?	Plot number 27/Number 27/27 ✓	1
17.	How many inside doors are in the building?	4/Four ✓	1
18.	How many 2 600 mm x 1 400 mm windows are in the building?	3/Three ✓	1
19.	Who checked the drawing of the new dwelling?	P Bot ✓	1
20.	Why will it be difficult to enter the dwelling at number 8 if the NGL is 300 mm lower than the FFL?	No step ✓ No porch/stoep No ramp The door is at a higher level	1
21.	Who was responsible for the printing of the building plan?	Frog printers ✓	1
22.	Deduce from the notes column the date that revision 1 took place.	14/04/2024 ✓	1
23.	In which town will the new dwelling be erected?	Malubu ✓	1
24.	Draw the symbol for a sink unit – double.		5
25.	Draw the symbol for hardcore filling.		2
26.	Calculate the metres of clear-view fencing that would be needed to fence plot number 23, excluding the driveway. Give your answer in metres and show ALL calculations.	$25\ 000\ \checkmark + 25\ 000\ \checkmark + 30\ 000\ \checkmark + 30\ 000\ \checkmark - 4\ 000\ \text{mm}\ \checkmark$ $= 106\ 000\ \text{mm} = 106\ \checkmark\ \text{m}$ OR $25 + 25 + 30 + 30 - 4 = 106\ \text{m}$	6
27.	The internal area of the garage is 39,6 m ² . Calculate the internal length of number 11. Give your answer in mm and show ALL calculations.	$39,6\ \text{m}^2\ \checkmark = 6,6\ \text{m}$ $6\ \text{m}\ \checkmark$ $= 6\ 600\ \text{mm}\ \checkmark$ OR $39\ 600\ 000\ \text{mm}^2$ $6\ 000\ \text{mm}$ $= 6\ 600\ \text{mm}$	3
		TOTAL:	40

QUESTION 3: CUPBOARDS, CASEMENTS, WALL-PANELLING AND QUANTITIES (SPECIFIC)

3.1 3.1.1 Kick plate ✓

3.1.2 Hardboard ✓

3.1.3 Hinges ✓

3.1.4 Screws ✓

3.1.5 Rail ✓

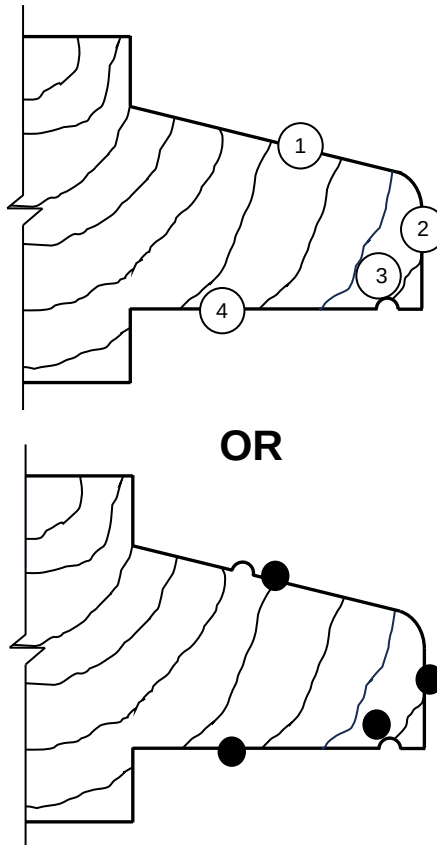
(5)

3.2

Member	Quantity	Unit	Length	Width	Thickness
Top	1	mm	<u>868</u> ✓	530	16
Base front rail	1	mm	868	<u>80</u> ✓	16
Shelf board	<u>4</u> ✓	mm	868	514	16
Table top	1	mm	<u>800</u> ✓	530	<u>32</u> ✓
Side of table	1	mm	<u>750</u> ✓	530	16
Back of cupboard	1	mm	<u>1 968</u> ✓	<u>868</u> ✓	16

(8)

3.3 3.3.1



NO.	ASSESSMENT CRITERIA	MARK
1	Bevel (at an angle)	1
2	Nose	1
3	Drip groove	1
4	Bottom	1
TOTAL:		4

(4)

3.3.2 Mistake in the casement:

- The windowpane is placed incorrectly ✓ in the bottom rail of the fanlight.
- The wooden bead/quarter round is placed incorrectly in the bottom rail of the fanlight.
- No hatching of the wooden bead/quarter round in the top rail of the casement.
- No drip groove on the bottom rail of the fanlight.

ANY ONE OF THE ABOVE

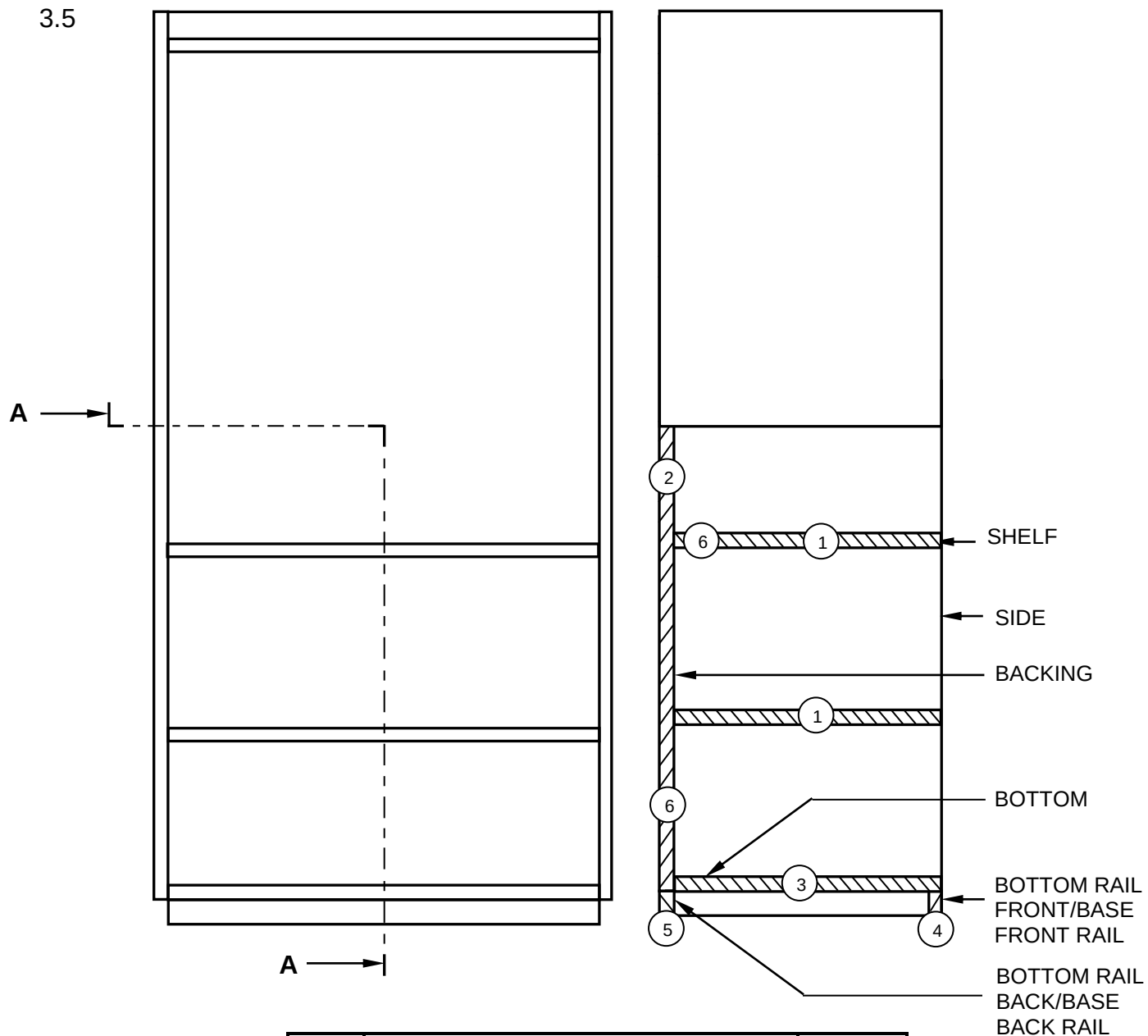
(1)

3.4 Constructing wall panelling:

1. **B** – Rough grounds ✓
2. **D** – Tongue and groove boards ✓
3. **C** – Skirting ✓
4. **A** – Quadrant mould ✓

(4)

3.5



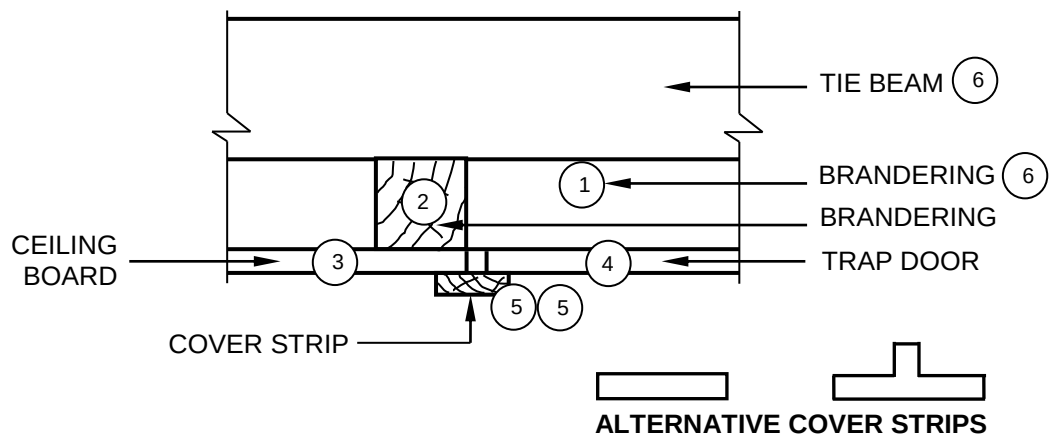
NO.	ASSESSMENT CRITERIA	MARK
1	Shelves	2
2	Backing	1
3	Bottom	1
4	Bottom rail (Front)	1
5	Bottom rail (Back)	1
6	Hatching (Correct parts)	2
TOTAL:		8

(8)
[30]

**QUESTION 4: ROOFS, CEILINGS, TOOLS, EQUIPMENT AND MATERIALS
(SPECIFIC)**

- | | | | |
|-----|-------|---|-----|
| 4.1 | 4.1.1 | F ✓ | (1) |
| | 4.1.2 | E ✓ | (1) |
| | 4.1.3 | J ✓ | (1) |
| | 4.1.4 | C ✓ | (1) |
| | 4.1.5 | B ✓ | (1) |
| | 4.1.6 | I ✓ | (1) |
| 4.2 | 4.2.1 | It provides a tough✓/durable✓/water-resistant/clear/
decorative/matte/sheen/glossy finish. | (2) |
| | 4.2.2 | <ul style="list-style-type: none">• The type of grading ✓• Mechanical grading• Visual grading ANY ONE OF THE ABOVE | (1) |
| | 4.2.3 | 60 minutes/1 hour ✓ | (1) |
| | 4.2.4 | <ul style="list-style-type: none">• 38 mm ✓• 50 mm ✓• 76 mm• 28 mm ANY TWO OF THE ABOVE | (2) |
| 4.3 | 4.3.1 | Cover strip/Wooden strip ✓ | (1) |

4.3.2



NO.	ASSESSMENT CRITERIA	MARK
1	Branding	1
2	Branding (Square)	1
3	Ceiling board	1
4	Trap door	1
5	Cover strip (any type of strip)	2
6	Any TWO labels	2
TOTAL:		8

(8)

4.4 Methods of fixing parts of a roof truss together:

- A – Gang nail/Nail plate ✓
- B – Bolt and nut ✓
- C – Screws/Nails ✓

(3)

4.5 Roof trusses:

- Lean-to roof truss ✓
- Coupled roof truss ✓
- Close couple roof truss ✓
- Collar-tie roof truss
- King post roof truss

ANY THREE OF THE ABOVE

(3)

4.6 Maximum spacing:

4.6.1 1 200 mm ✓

(1)

4.6.2 345 mm ✓

(1)

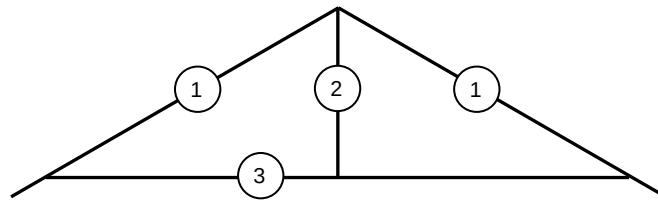
4.6.3 300 mm ✓

(1)

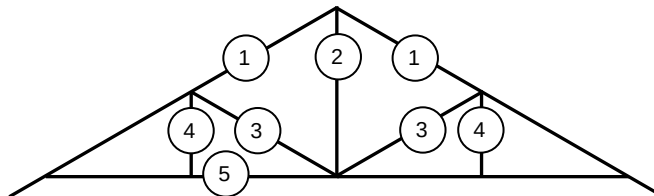
- 4.7 Reasons for using a closed eave:
- Provide a more attractive finish ✓
 - Prevent birds/insects/vermin from nesting in the roof spacing ✓
 - Beam filling is not needed ✓
 - Prevent wind from lifting the roof covering
- ANY THREE OF THE ABOVE** (3)
- 4.8
- A** – Hip/Corner rafter ✓
 - B** – Jack rafter/Jack truss ✓
 - C** – Wall plate ✓
 - D** – Fascia board ✓
- (4)
- 4.9 Members of a Collar-tie roof truss:
- Ridge beam ✓
 - Rafter ✓
 - Collar beam ✓
- (3)
[40]

QUESTION 5: CENTERING, FORMWORK, SHORING AND GRAPHICS AS MEANS OF COMMUNICATION (SPECIFIC)

5.1

**KING POST ROOF TRUSS**

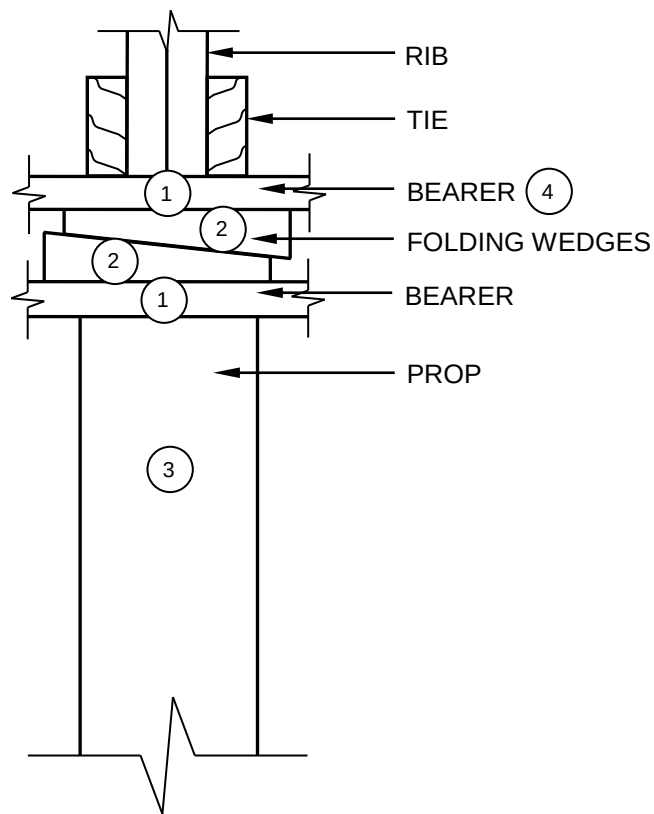
NO.	ASSESSMENT CRITERIA	MARK
1	Rafters (Must include overhang)	2
2	King post	1
3	Tie beam	1
	TOTAL:	4

**SA HOWE ROOF TRUSS**

NO.	ASSESSMENT CRITERIA	MARK
1	Rafters (Must include overhang)	2
2	King post	1
3	Struts	2
4	Queen posts	2
5	Tie beam	1
	TOTAL:	8

(12)

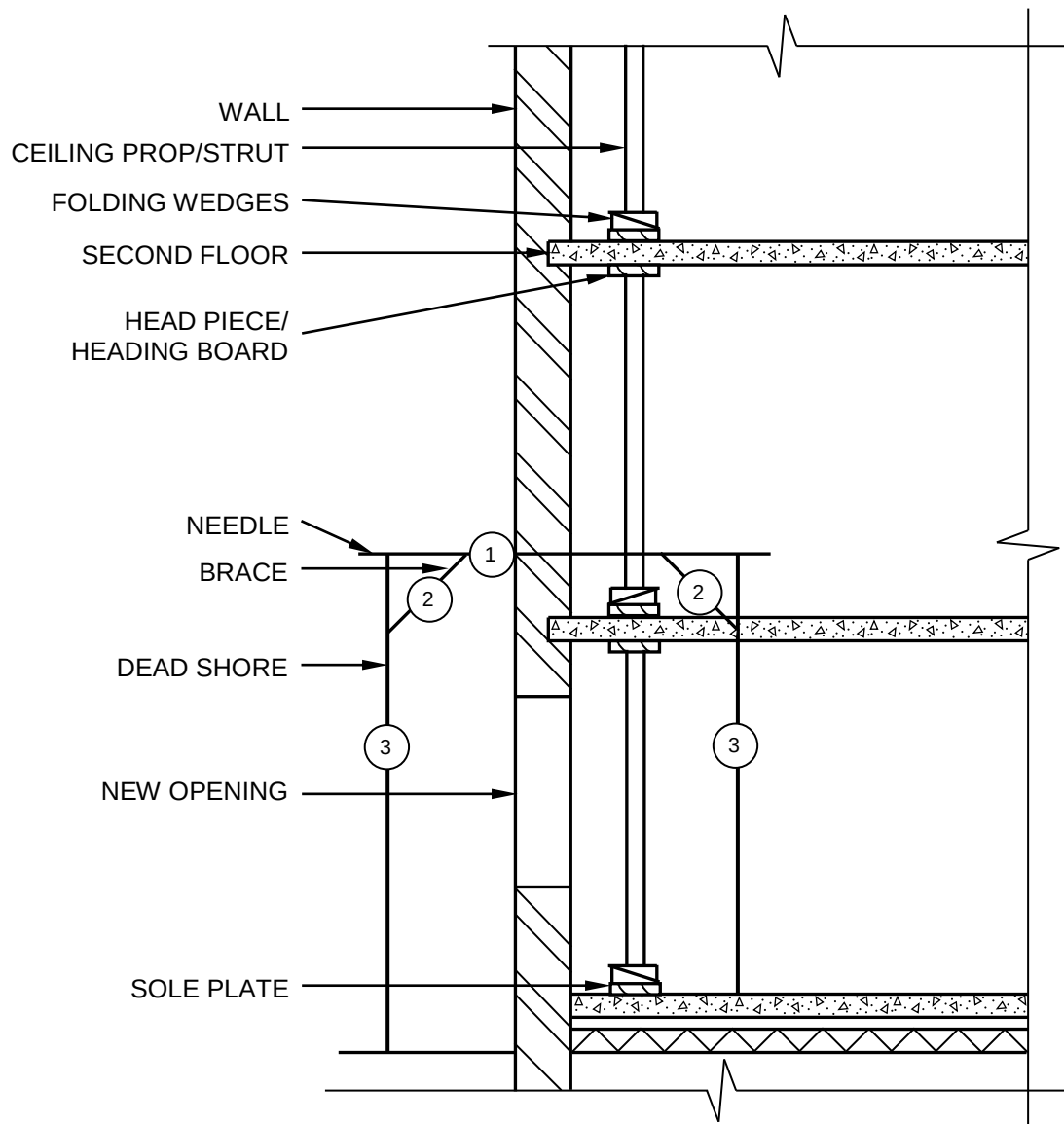
5.2



NO.	ASSESSMENT CRITERIA	MARK
1	Bearers	2
2	Folding wedges	2
3	Prop	1
4	Any ONE label	1
	TOTAL:	6

(6)

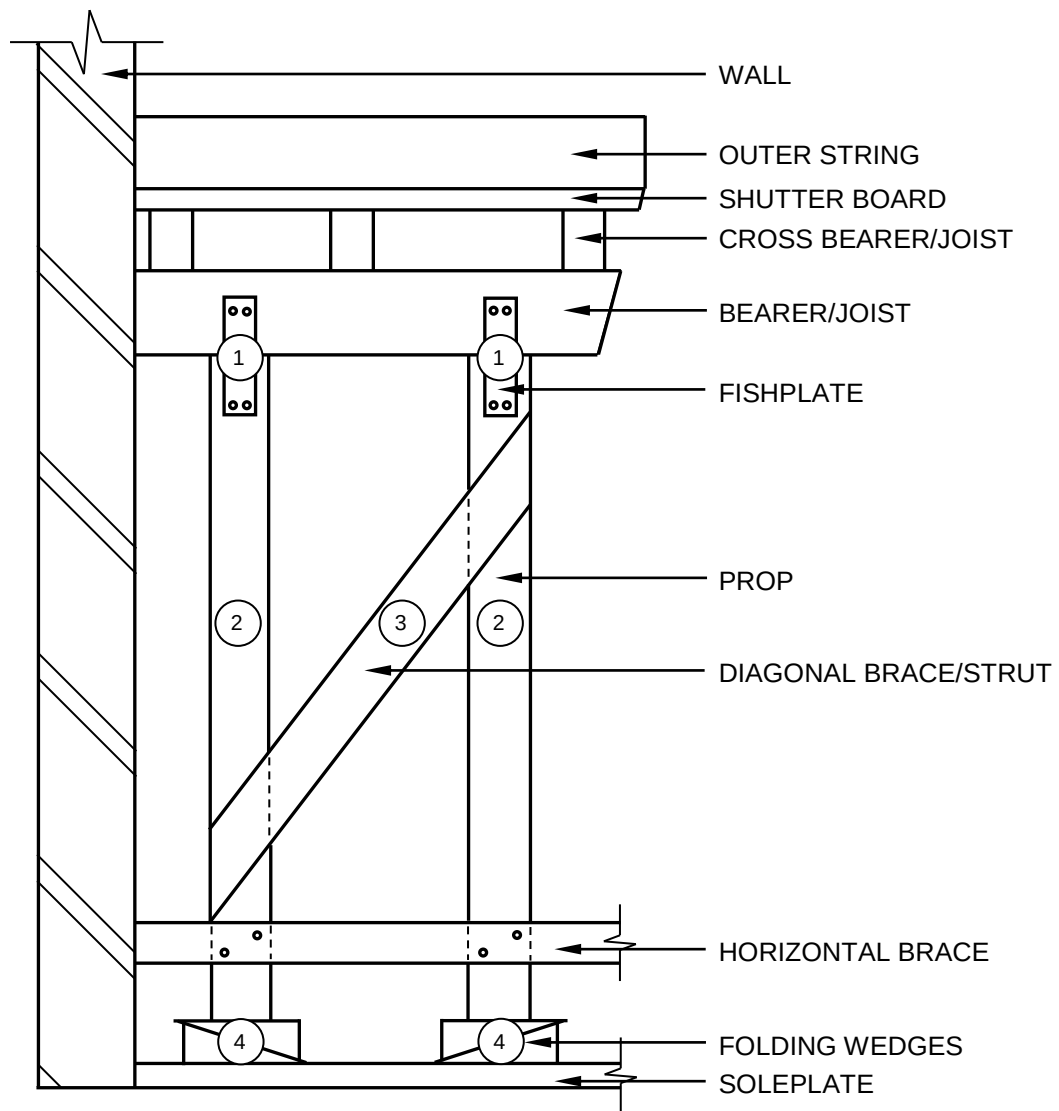
5.3



NO.	ASSESSMENT CRITERIA	MARK
1	Needle	1
2	Braces	2
3	Dead shores	2
	TOTAL:	5

(5)

5.4



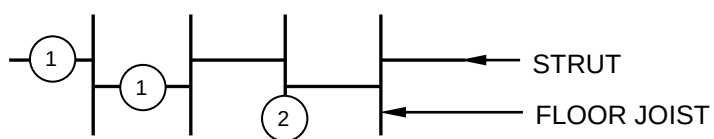
NO.	ASSESSMENT CRITERIA	MARK
1	Fishplates	2
2	Props	2
3	Diagonal brace/Strut	1
4	Folding wedges	2
TOTAL:		7

(7)
[30]

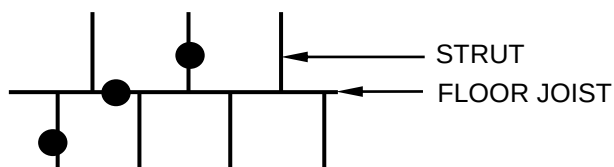
QUESTION 6: SUSPENDED FLOORS, STAIRCASES, IRONMONGERY, DOORS AND JOINING (SPECIFIC)

- | | | | |
|-----|-------|---|-----|
| 6.1 | 6.1.1 | D ✓ | (1) |
| | 6.1.2 | A ✓ | (1) |
| | 6.1.3 | C ✓ | (1) |
| | 6.1.4 | B ✓ | (1) |
| | 6.1.5 | C ✓ | (1) |
| 6.2 | 6.2.1 | A door is fixed to a doorframe by using various types of hinges ✓
that are screwed onto the frame and the door. ✓ | (2) |
| | 6.2.2 | <ul style="list-style-type: none">• A galvanised strap/roof wire is built into/fixed onto the wall ✓
and strapped over the truss. ✓• If the wall plate is fixed to the wall, the galvanised strap can be
nailed to the wall plate and strapped over the truss. ANY ONE OF THE ABOVE | (2) |

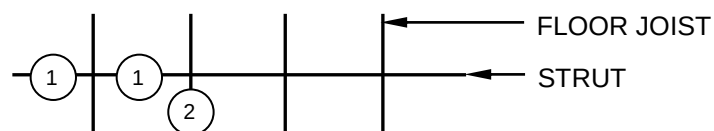
6.3



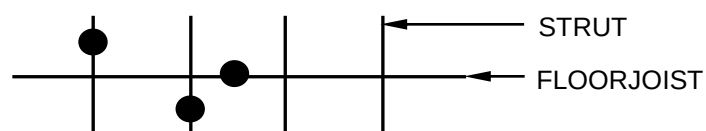
STAGGERED STRUTS
OR



NO.	ASSESSMENT CRITERIA	MARK
1	Staggered struts	2
2	Floor joists	1
	TOTAL:	3



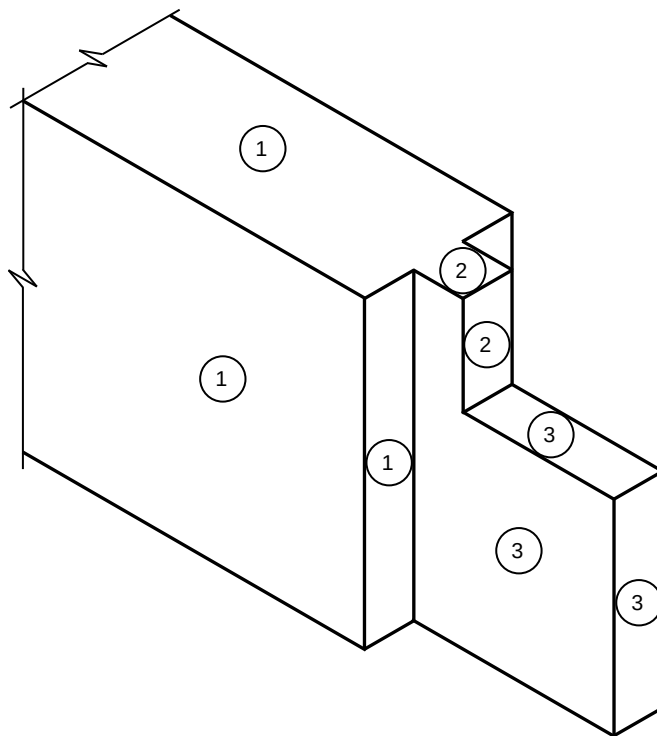
INLINE STRUTS
OR



NO.	ASSESSMENT CRITERIA	MARK
1	In-line struts	2
2	Floor joists	1
	TOTAL:	3

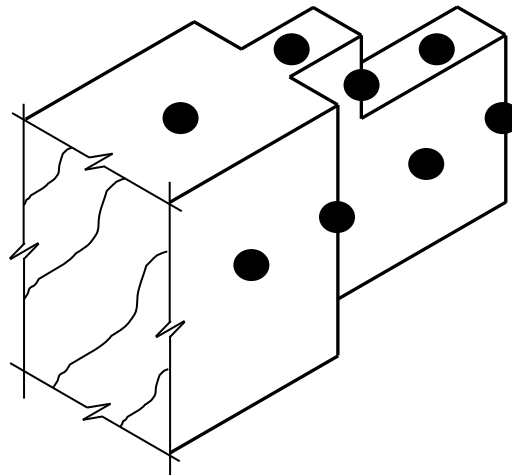
(6)

6.4



**NOTE: DRAWING
MAY BE ROTATED**

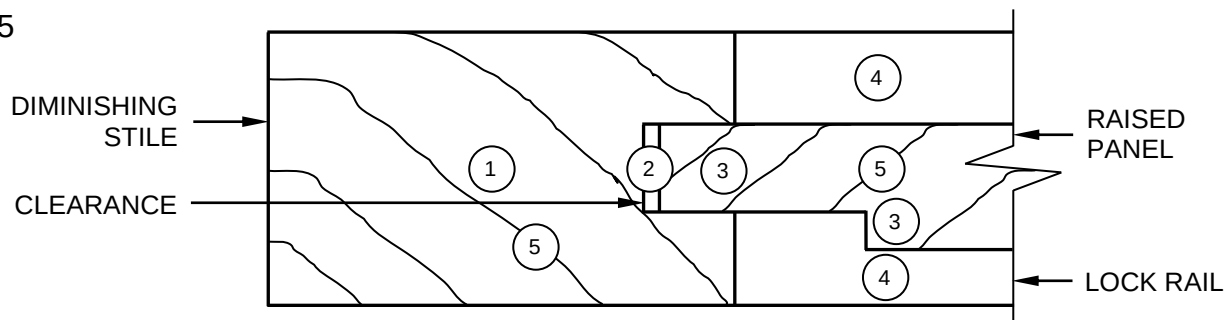
OR



NO.	ASSESSMENT CRITERIA	MARK
1	Top rail surfaces	3
2	Haunch	2
3	Tenon surfaces	3
	TOTAL:	8

(8)

6.5



NO.	ASSESSMENT CRITERIA	MARK
1	Diminishing stile	1
2	Clearance	1
3	Raised panel	2
4	Lock rail	2
5	Hatching (Correct parts)	2
TOTAL:		8

(8)

6.6

A - Handrail ✓

(1)

B - Half landing ✓

(1)

C - Outer stringer ✓

(1)

D - Riser ✓

(1)

E - Tread ✓

(1)

F - Nosing of tread ✓

(1)

G - Newel post ✓

(1)

H - Baluster ✓

(1)

I - Wall stringer ✓

(1)

[40]**TOTAL: 200**